CHAPTER 5

BREATHING AND EXCHANGE OF GASES

CHAPTER ESSENCE AT A GLANCE:
Thoracic mechanics of pulmonary ventilation, partial pressure gradients, and hemoglobin saturation dynamics governing gas transport.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Muscular Mechanics of Pulmonary Ventilation:** Inspiration is an active process driven by contraction of the diaphragm (flattens downward) and external intercostal muscles (elevates ribs and sternum), expanding thoracic volume, lowering intrapulmonary pressure below atmospheric pressure to draw air into lungs. Expiration during rest is passive as muscles relax, reducing thoracic volume and elevating pressure above atmospheric level.

• **Respiratory Volumes and Diagnostic Capacities:** Standard spirometry: Tidal Volume (TV = 500 mL), Inspiratory Reserve Volume (IRV = 2500 to 3000 mL), Expiratory Reserve Volume (ERV = 1000 to 1100 mL), and Residual Volume (RV = 1100 to 1200 mL). Vital Capacity ($VC = ERV + TV + IRV ■pprox 4600 ext{ mL}$) represents maximum air exhaled after maximal inspiration; Total Lung Capacity ($TLC = VC + RV ■pprox 5800 ext{ mL}$).

• **Alveolar-Capillary Gas Diffusion Barrier:** Gas exchange occurs across a 3-layered diffusion membrane thinner than 1 mm: squamous epithelium of alveoli, endothelium of alveolar capillaries, and intervening basement membrane. Driven by partial pressure gradients: alveolar pO2 (104 mm Hg) exceeds deoxygenated blood pO2 (40 mm Hg); blood pCO2 (45 mm Hg) exceeds alveolar pCO2 (40 mm Hg). Solubility of CO2 is 20 to 25 times higher than O2.

• **Gas Transport and Dissociation Kinetics:** Oxygen transport: 97% as oxyhemoglobin, 3% dissolved in plasma. Oxygen dissociation curve is sigmoid; shifts right (Bohr effect) during low pO2, high pCO2, high H+ acidity, and elevated temperature (favoring oxygen unloading at tissues). Carbon dioxide transport: 70% as bicarbonate ions (HCO3- facilitated by erythrocyte carbonic anhydrase), 20 to 25% as carbamino-hemoglobin, and 7% dissolved in plasma.

ESSENTIAL PHYSIOLOGICAL PATHWAYS & STRUCTURES

$$VC = ERV + TV + IRV \approx 4600 \text{ mL}, \quad TLC = VC + RV \approx 5800 \text{ mL}$$

Formulation of Vital Capacity and Total Lung Capacity

$$CO_2 + H_2O \xrightleftharpoons{\text{Carbonic Anhydrase}} H_2CO_3 \rightleftharpoons HCO_3^- + H^+$$

Erythrocyte Carbonic Anhydrase Mediated Bicarbonate Cascade

$$O_2 \text{ Tissue Delivery} = 100 \text{ mL oxygenated blood} \longrightarrow 5 \text{ mL } O_2 \text{ delivered to tissues}$$

Physiological Oxygen Delivery Yield under Normal Resting Conditions

$$CO_2 \text{ Alveolar Release} = 100 \text{ mL deoxygenated blood} \longrightarrow 4 \text{ mL } CO_2 \text{ released into alveoli}$$

Physiological Carbon Dioxide Clearance Yield per Circulation Cycle

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** The oxygen-hemoglobin dissociation curve shifts to the RIGHT (promoting oxygen release) under high pCO2, low pH, and high temperature at tissue level.

• **Alert:** Every 100 mL of oxygenated blood delivers approximately 5 mL of O2 to active tissues; every 100 mL of deoxygenated blood delivers approximately 4 mL of CO2 to alveoli.

• **Alert:** Emphysema is a chronic respiratory disorder characterized by alveolar wall destruction reducing the gas exchange surface, primarily caused by cigarette smoking.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Define Vital Capacity (VC) and Residual Volume (RV). State their normal approximate volumes in a healthy adult. [Score: 2 Marks]

Answer & Key Valuation Steps:

• Vital Capacity (VC): The maximum volume of air a person can breathe out after a forced inspiration (VC = ERV + TV + IRV), approximately 4000 to 4600 mL. **[1 Mark]**

• Residual Volume (RV): The volume of air remaining in the lungs even after a forceful expiration, approximately 1100 to 1200 mL. **[1 Mark]**

2. What is emphysema? State its primary causative factor and the specific pathological damage caused to the lungs. [Score: 2 Marks]

Answer & Key Valuation Steps:

• Definition and cause: Emphysema is a chronic respiratory disorder majorly caused by cigarette smoking. **[1 Mark]**

• Pathological damage: Alveolar walls are progressively damaged and broken down, severely reducing the total respiratory surface area available for gas exchange. **[1 Mark]**

3. Describe the coordinated muscular contractions of the diaphragm and intercostal muscles during normal inspiration and expiration in humans. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Inspiration mechanism: Initiated by contraction of the diaphragm, which flattens and increases thoracic cavity volume along the antero-posterior axis. **[1 Mark]**

• Intercostal muscles: Contraction of external intercostal muscles lifts the ribs and sternum, increasing thoracic volume along the dorso-ventral axis; pulmonary volume expands, intrapulmonary pressure drops below atmospheric pressure, and air rushes in. **[1 Mark]**

• Expiration mechanism: Relaxation of the diaphragm and external intercostal muscles returns them to their dome shape and resting positions, compressing the thorax, raising pulmonary pressure, and expelling air. **[1 Mark]**

4. Explain the role of the enzyme carbonic anhydrase in facilitating the transport of carbon dioxide as bicarbonate ions in blood. [Score: 3 Marks]

Answer & Key Valuation Steps:

• RBCs contain high concentrations of the enzyme carbonic anhydrase, with minute quantities present in plasma. **[1 Mark]**

• At tissue level where pCO_2 is high, carbonic anhydrase catalyzes $CO_2 + H_2O$ to form carbonic acid (H_2CO_3), which dissociates into bicarbonate (HCO_3^-) and H^+ ions. **[1 Mark]**

• At alveolar level where pCO_2 is low, the reaction is reversed: bicarbonate ions recombine with H^+ to release CO_2 and H_2O, facilitating rapid CO_2 diffusion into alveoli (70% of CO_2 is transported this way). **[1 Mark]**

5. Describe the characteristic shape of the oxygen-hemoglobin dissociation curve. List three tissue conditions that shift this curve to the right. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Curve shape: A standard plot of percentage saturation of hemoglobin with oxygen against partial pressure of oxygen (pO_2) produces a characteristic sigmoid (S-shaped) curve. **[1 Mark]**

• Tissue conditions shifting curve to the right (Bohr effect, promoting O_2 unloading): (1) Low partial pressure of oxygen (pO_2), (2) High partial pressure of carbon dioxide (pCO_2), (3) High hydrogen ion concentration (low pH and acidosis) and elevated body temperature. **[2 Marks]**

6. Name the three cellular layers that constitute the respiratory diffusion membrane between pulmonary alveoli and capillaries. **[Score: 3 Marks]**

Answer & Key Valuation Steps:

• Layer 1: The thin squamous epithelium of the alveolar wall. **[1 Mark]**

• Layer 2: The single-layered endothelial lining of the alveolar capillary wall. **[1 Mark]**

• Layer 3: The thin non-cellular basement membrane situated between the alveolar epithelium and capillary endothelium (total thickness is less than 0.5 mm). **[1 Mark]**

7. A lowlander ascends rapidly to an altitude of 4200 meters and experiences dizziness, headache, nausea, and shortness of breath (altitude sickness). Explain the physiological cause of these symptoms and describe three compensatory adaptations the body makes over time. **[Score: 4 Marks]**

Answer & Key Valuation Steps:

• Physiological cause: At high altitudes, atmospheric pressure is low, so the partial pressure of oxygen (pO2) is reduced; consequently, hemoglobin cannot achieve sufficient oxygen saturation, leading to tissue hypoxia. **[1.5 Marks]**

• Compensatory adaptation 1: Kidneys release erythropoietin hormone, which stimulates bone marrow to accelerate RBC production (polycythemia), elevating oxygen-carrying capacity. **[1 Mark]**

• Compensatory adaptation 2: The respiratory center increases breathing rate (hyperventilation) to enhance alveolar oxygen intake. **[0.75 Mark]**

• Compensatory adaptation 3: Hemoglobin decreases its oxygen-binding affinity (increased 2,3-BPG), facilitating easier release of oxygen to deprived tissues. **[0.75 Mark]**